

Waiz Khan

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EDUCATION

Johns Hopkins University

M.S.E. in Data Science

Baltimore, MD

Aug. 2024 – Dec. 2026

- Coursework: Investment Science, Financial Derivatives, Deep Learning, Artificial Intelligence, Statistical Learning

GD Goenka University

B.Tech. in Computer Science & Engineering

Gurugram, India

Aug. 2019 – Nov. 2023

- Coursework: Operations Research, Numerical Analysis, Stochastic Processes, Linear Algebra, Data Mining, Distributed Systems

EXPERIENCE

Research Assistant

Johns Hopkins University, Center for Language and Speech Processing

Dec. 2025 – Present

Baltimore, MD

Advisor: Dr. Philipp Koehn

- Developing a multimodal neural machine translation system for ancient Egyptian hieroglyphs, engineering hybrid encoder architectures to fuse visual representations with symbolic sequences in extreme low-resource settings
- Designing custom preprocessing pipelines for high-noise domains; benchmarking representation learning strategies across vision and sequence models to address data scarcity and improve translation quality

Research Assistant

GD Goenka University

Aug. 2022 – Nov. 2022

Gurugram, India

Advisor: Dr. Yogesh Kumar

- Built end-to-end ML, causal inference, and optimization pipeline for educational risk prediction on 47,027 records, achieving 84.1% ROC-AUC with DoWhy causal validation
- Developed ILP allocator that doubled expected impact per budget dollar; stress-tested with 10,000 Monte Carlo simulations and VaR reporting

Data Scientist Intern

EBMS Solutions Pvt Ltd

May 2022 – Sep. 2022

Chennai, India

- Engineered churn prediction pipeline using XGBoost with time-series CV; deployed to production CRM with automated retraining; achieved +15% retention lift and AUC 0.82
- Built self-serve analytics layer with curated cohort tables and data quality monitoring; reduced reporting cycle time by 30%

PROJECTS

Black-Scholes Options Pricer | *Python, NumPy, SciPy, Streamlit*

Jan. 2025

- Built production-grade European options pricer with Greeks calculation (Delta, Gamma, Vega, Theta, Rho) and implied volatility solver using Newton-Raphson with Brent fallback
- Implemented arbitrage diagnostics (put-call parity, butterfly spreads, strike monotonicity); 91+ tests, 85% coverage

Monte Carlo Risk Engine | *Python, pandas, Plotly, Streamlit*

Jan. 2025

- Developed portfolio risk simulator computing VaR, CVaR, Sharpe, Sortino, and Max Drawdown using multivariate normal and GBM models
- Applied Ledoit-Wolf covariance shrinkage and walk-forward validation to prevent look-ahead bias; 90%+ test coverage

TECHNICAL SKILLS

Languages: Python, R, SQL, Rust, Swift, Java, C++, JavaScript, Bash

ML/AI: PyTorch, TensorFlow, scikit-learn, XGBoost; CNNs, transformers, NLP; causal inference (DoWhy); time-series CV

Quant/Finance: Black-Scholes, Greeks, VaR/CVaR/ES; Monte Carlo simulation; portfolio optimization; backtesting

Data/Cloud: pandas, NumPy, SciPy; PostgreSQL, Spark; AWS (EC2, S3, SageMaker); Git, Docker; Tableau, Power BI

Security/iOS: Digital forensics, memory/log analysis, incident response; SwiftUI, Core ML, HealthKit